# My Solutions to Problem Sheet 2 QFT by David Tong

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## 1.

According to the definitions made:

$$\begin{aligned}[a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0\end{aligned} \tag{1}$$

$$\begin{aligned}[a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0\end{aligned} \tag{2}$$

$$\begin{aligned}[a_n^\dagger, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] - i[p_n, q_m]\} = 0\end{aligned} \tag{3}$$

$$\begin{aligned}[a_n, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] + i[p_n, q_m]\} = \delta_{nm}\end{aligned} \tag{4}$$

Using 4:

$$\begin{aligned}H &= \sum_n \frac{1}{2}\omega_n(a_n a_n^\dagger + a_n^\dagger a_n) = \sum_n \omega_n(a_n^\dagger a_n + \frac{1}{2}) \\ &= \sum_n \omega_n((\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n)(\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n) + \frac{1}{2}) \\ &= \sum_n \omega_n(\frac{1}{2}\omega_n q_n^2 + \frac{1}{2\omega_n}p_n^2 + \frac{i}{2}[q_n, p_n] + \frac{1}{2}) \\ &= \sum_n \frac{1}{2}(p_n^2 + \omega_n^2 q_n^2)\end{aligned} \tag{5}$$

If we use the first line of the above calculation as $H$:

$$H|0\rangle = \sum_n \omega_n (a_n^\dagger a_n + \frac{1}{2})|0\rangle = \infty |0\rangle \tag{6}$$

In the above $H$ terms, the only term that produces a non-zero energy when acting on vacuum $|0\rangle$ is the constant $\frac{1}{2}$ term, by removing it:

$$H|0\rangle = \sum_n \omega_n a_n^\dagger a_n |0\rangle = 0|0\rangle \tag{7}$$

$$\begin{aligned}
[H, a_n^\dagger] &= [\sum_{m=1}^{\infty} \omega_m a_m^\dagger a_m, a_n^\dagger] = \sum_{m=1}^{\infty} \omega_m [a_m^\dagger a_m, a_n^\dagger] \\
&= \sum_{m=1}^{\infty} \omega_m a_m^\dagger [a_m, a_n^\dagger] = \omega_n a_n^\dagger
\end{aligned} \tag{8}$$

Therefore:

$$\begin{aligned}
g_l \equiv [H, (a_n^\dagger)^l] = [H, a_n^\dagger](a_n^\dagger)^{l-1} + a_n^\dagger[H, (a_n^\dagger)^{l-1}] &= \omega_n (a_n^\dagger)^l + a_n^\dagger [H, (a_n^\dagger)^{l-1}] \\
= \omega_n (a_n^\dagger)^l + a_n^\dagger g_{l-1} &= \omega_n (a_n^\dagger)^l + a_n^\dagger (\omega_n)
\end{aligned} \tag{9}$$

$$H|l_1, l_2, ..., l_N\rangle = H(a_1^\dagger)^{l_1}(a_2^\dagger)^{l_2}...(a_N^\dagger)^{l_N}|0\rangle \\ = \tag{10}$$